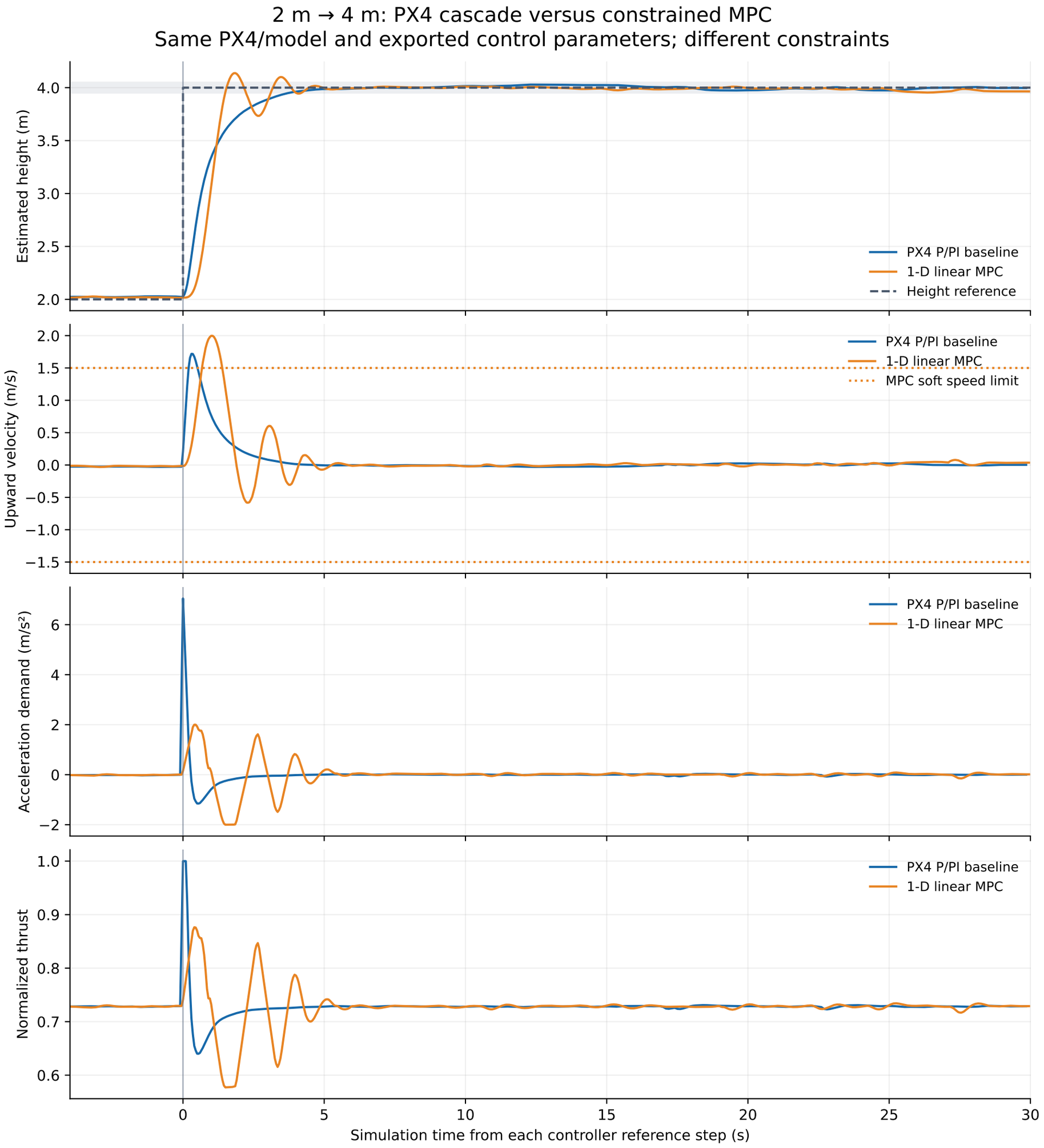
2 m → 4 m: PX4 cascade versus constrained MPC
Same PX4/model and exported control parameters; different constraints
Estimated height (m)
PX4 P/PI baseline
1-D linear MPC
Height reference
Upward velocity (m/s)
MPC soft speed limit
Acceleration demand (m/s²)
Normalized thrust
Simulation time from each controller reference step (s)